

3_Prepare_Test_Plan → Test_Plan

TEST PLAN DOCUMENT

Project Title

Data-Driven Testing of SauceDemo Login Functionality Using Selenium WebDriver, TestNG, and CSV File

1. Document Information

Item	Details
Project Name	SauceDemo Login Testing
Testing Type	Manual Testing and Automation Testing
Application Under Test	SauceDemo
Module	Login Module
Prepared By	QA Tester
Test Environment	Web Application
Automation Tool	Selenium WebDriver
Testing Framework	TestNG
Programming Language	Java
Data Source	CSV File
Build Tool	Maven

2. Introduction

This Test Plan document defines the overall testing strategy, objectives, scope, resources, schedule, and approach for testing the Login functionality of the SauceDemo application.

The purpose of this document is to ensure that all testing activities are properly planned and executed in a structured manner. The Login functionality is a critical module because it controls user authentication and access to the application.

This project includes both Manual Testing and Automation Testing activities. Manual testing is performed to validate requirements and create test documentation, while Automation Testing is implemented using Selenium WebDriver, Java, TestNG, and CSV-based Data-Driven Testing.

3. Project Overview

The SauceDemo application is a sample e-commerce web application used for testing and learning purposes.

The Login Module allows users to authenticate themselves using valid credentials and access the application.

The project aims to verify that:

- Valid users can successfully log in.
 - Invalid users cannot access the application.
 - Appropriate validation messages are displayed.
 - Multiple login combinations can be executed using Data-Driven Testing.
 - Automation scripts execute reliably and generate reports.
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4. Test Objectives

The primary objectives of testing are:

Objective 1

Verify successful login using valid credentials.

Objective 2

Verify unsuccessful login using invalid credentials.

Objective 3

Validate mandatory field requirements.

Objective 4

Verify error message functionality.

Objective 5

Ensure successful redirection to the Products page.

Objective 6

Execute login tests using multiple data combinations.

Objective 7

Implement Data-Driven Testing using CSV files.

Objective 8

Generate execution reports and screenshots.

Objective 9

Identify and report defects.

Objective 10

Ensure complete test coverage of the Login functionality.

5. Scope of Testing

In Scope

The following functionalities are included in testing:

Login Page Validation

- Username Field
- Password Field
- Login Button
- Error Messages

Positive Testing

- Login using valid credentials
- Login using approved test users

Negative Testing

- Invalid Username
- Invalid Password

- Blank Username
- Blank Password
- Blank Credentials
- Locked User Login

Manual Testing Activities

- Requirement Analysis
- Test Planning
- Test Scenario Creation
- Test Case Creation
- Test Execution
- Defect Reporting

Automation Testing Activities

- Selenium Script Development
- Data-Driven Testing
- CSV Integration
- TestNG Execution
- Report Generation

Out of Scope

The following functionalities are excluded:

- Product Module Testing
- Shopping Cart Testing
- Checkout Functionality
- Payment Testing
- Performance Testing

- Security Testing
 - API Testing
 - Mobile Application Testing
 - Database Testing
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6. Testing Strategy

The project follows a combination of Manual Testing and Automation Testing.

Manual Testing Strategy

The following activities will be performed:

Requirement Analysis

Understanding business and functional requirements.

Test Planning

Preparing testing strategy and schedules.

Test Scenario Design

Identifying high-level test conditions.

Test Case Design

Creating detailed test cases.

Test Execution

Executing test cases manually.

Defect Reporting

Reporting identified issues.

Test Closure

Preparing final test summary.

Automation Testing Strategy

Automation testing will be implemented using:

- Selenium WebDriver
- Java
- TestNG
- Maven
- CSV File

The automation framework will:

- Read test data from CSV files.
 - Execute multiple login combinations.
 - Validate expected outcomes.
 - Capture screenshots.
 - Generate reports.
 - Support easy maintenance.
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7. Test Approach

The testing process will follow the Software Testing Life Cycle (STLC).

Phase 1

Requirement Analysis

Phase 2

Test Planning

Phase 3

Test Scenario Creation

Phase 4

Test Case Development

Phase 5

Test Data Preparation

Phase 6

Test Execution

Phase 7

Defect Reporting

Phase 8

Automation Execution

Phase 9

Test Closure

8. Test Environment

Testing will be conducted using the following environment.

Hardware Requirements

Processor:
Intel i3 or above

RAM:
8 GB or above

Storage:
20 GB Free Space

Internet:
Stable Internet Connection

Software Requirements

Operating System:
Windows 10 / Windows 11

Browser:
Google Chrome

JDK:
Java 21

IDE:
Eclipse IDE

Automation Tool:
Selenium WebDriver

Testing Framework:
TestNG

Build Tool:
Maven

Version Control:
GitHub

Reporting Tool:
Extent Reports

9. Test Data Requirements

Test data will be maintained in CSV format.

The following types of data will be used:

Positive Data

standard_user

secret_sauce

Expected Result:
Login Successful

Negative Data

Invalid Username

Invalid Password

Blank Username

Blank Password

Locked User

Expected Result:
Error Message

10. Entry Criteria

Testing can begin only when:

- Requirements are available.
 - Login functionality is accessible.
 - Test environment is ready.
 - Test data is prepared.
 - Browser setup is completed.
 - Selenium framework is configured.
 - Required tools are installed.
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11. Exit Criteria

Testing will be considered complete when:

- All planned test cases are executed.
- Critical defects are resolved.
- Test reports are generated.
- Test coverage is achieved.

- Automation scripts execute successfully.
 - Test closure report is prepared.
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12. Test Deliverables

The following deliverables will be produced during the project:

Documentation Deliverables

- Requirement Analysis Document
 - Test Plan Document
 - Test Scenarios
 - Test Cases
 - RTM
 - Test Execution Report
 - Defect Report
 - Test Closure Report
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Automation Deliverables

- Selenium Scripts
- CSV Test Data
- Config File
- TestNG XML
- Maven Project
- Extent Report
- TestNG Report
- Screenshots

13. Roles and Responsibilities

QA Tester

Responsibilities:

- Analyze requirements.
 - Prepare test documents.
 - Create test cases.
 - Execute tests.
 - Report defects.
 - Develop automation scripts.
 - Generate reports.
 - Prepare closure report.
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14. Risk Analysis

Risk 1

Application Unavailability

Impact:

Testing delays

Mitigation:

Retest after application availability.

Risk 2

Requirement Changes

Impact:

Rework of test cases.

Mitigation:
Maintain updated documentation.

Risk 3

Environment Issues

Impact:
Execution failure.

Mitigation:
Validate environment before execution.

Risk 4

Invalid Test Data

Impact:
Incorrect results.

Mitigation:
Review and validate test data.

15. Defect Management Process

The following defect lifecycle will be followed:

New

↓

Assigned

↓

Open

↓

Fixed

↓

Retest



Verified



Closed

If the defect still exists:

Retest



Reopen



Assigned Again

16. Test Metrics

The following metrics will be collected:

- Total Test Cases
 - Executed Test Cases
 - Passed Test Cases
 - Failed Test Cases
 - Blocked Test Cases
 - Defect Count
 - Defect Density
 - Pass Percentage
 - Fail Percentage
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17. Success Criteria

The project will be considered successful when:

- Login functionality works as expected.
 - All critical scenarios are tested.
 - Defects are identified and reported.
 - Automation scripts execute successfully.
 - Reports are generated correctly.
 - Test coverage is achieved.
 - Stakeholders approve test results.
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18. Conclusion

This Test Plan provides a structured approach for testing the SauceDemo Login functionality using both Manual Testing and Automation Testing techniques. The document defines the testing scope, objectives, strategy, resources, schedule, risks, deliverables, and success criteria. Following this plan will ensure systematic testing, improved quality, better defect detection, and successful implementation of Data-Driven Automation Testing.